[Logo] `logo.svg`

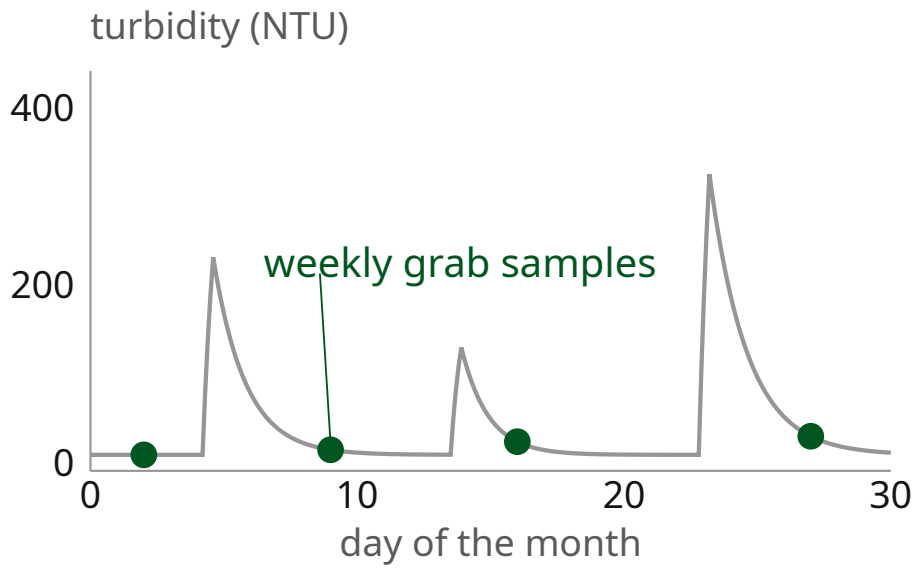
A low-cost optical sensor for storm-driven river turbidity

Continuous readings within 5 % of a laboratory instrument

[Presenter name]

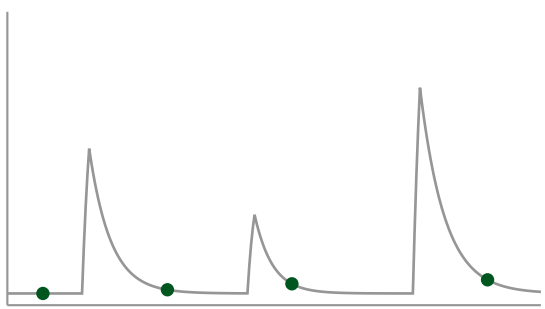
[Department] · [University]

[Conference] · [Date]

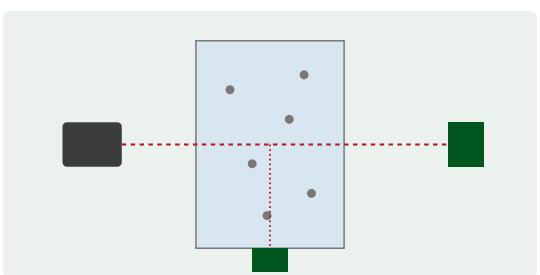


Outline

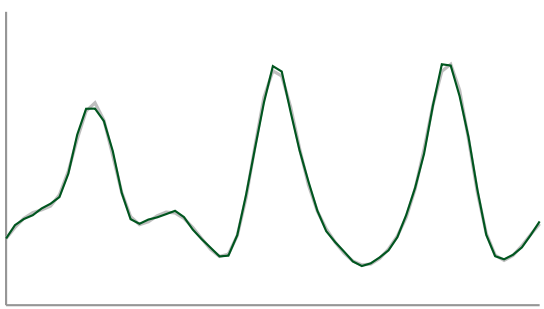
[Logo] logo.svg



1. Need

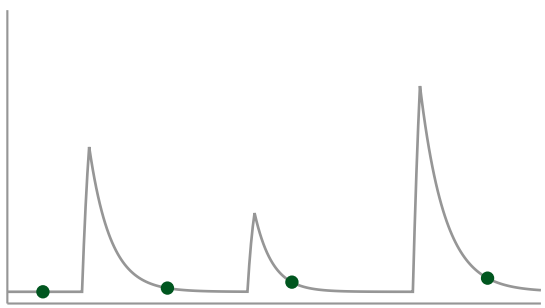


2. Sensor

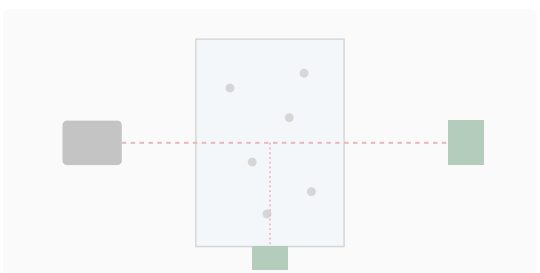


3. Field test

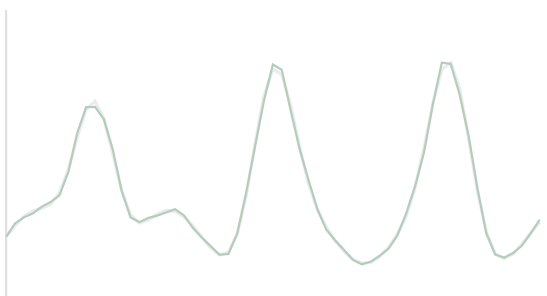
[Logo] logo.svg



1. Need



2. Sensor

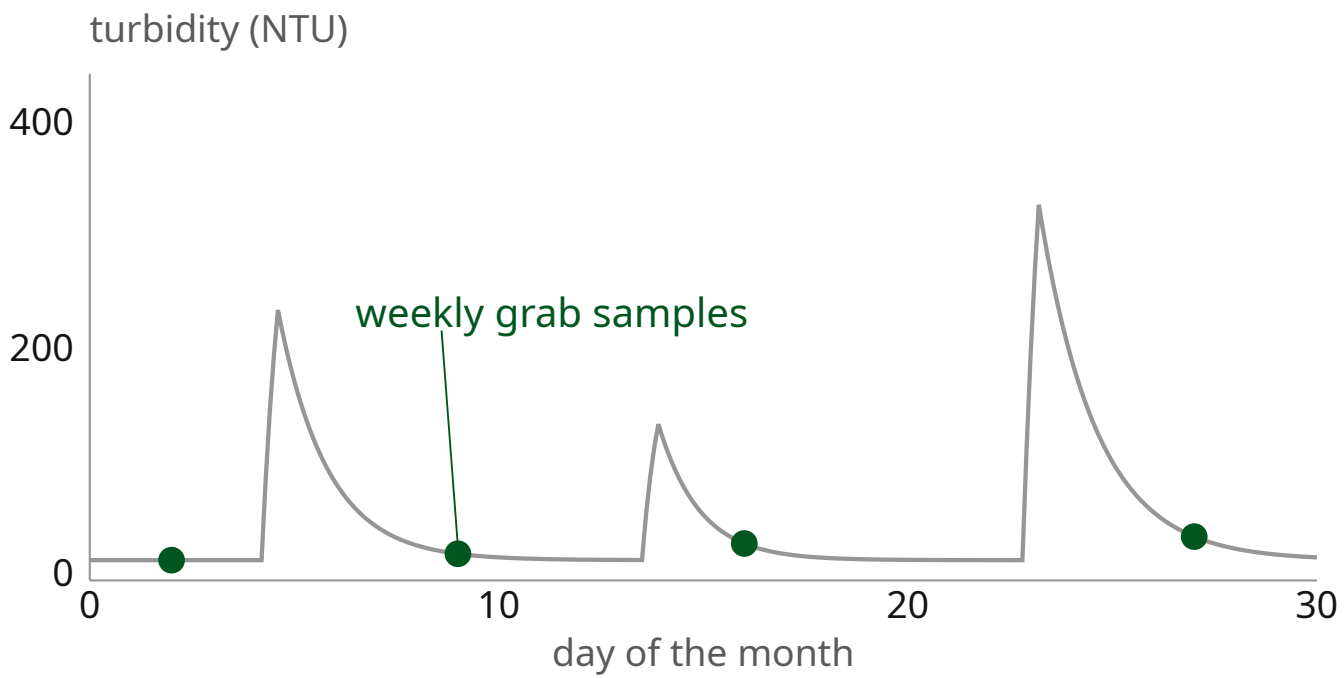


3. Field test

[Logo] logo.svg

Storm loads

Storms carry most of the sediment, yet weekly samples miss them

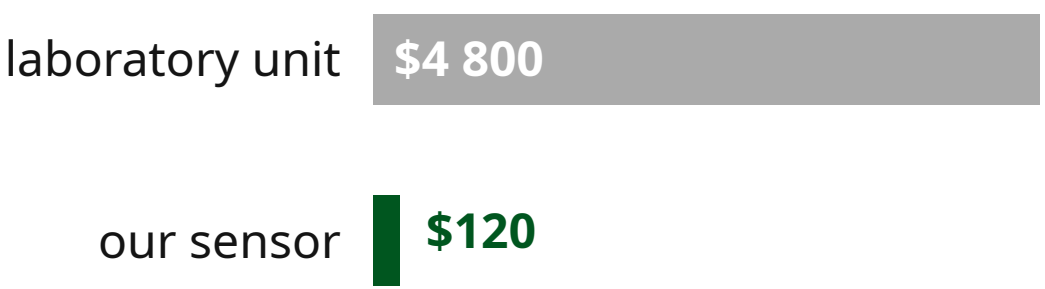


Illustrative record; sediment share after [Author, year].

Cost

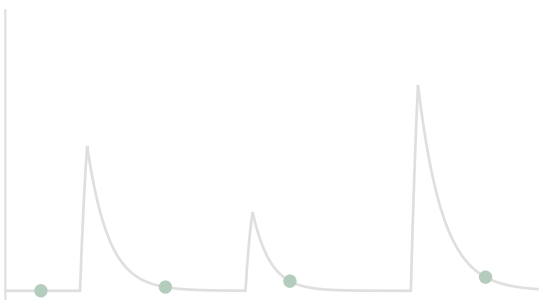
[Logo] logo.svg

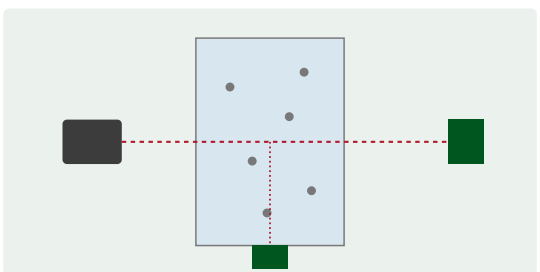
A laboratory turbidimeter costs about 40 times the parts of our sensor

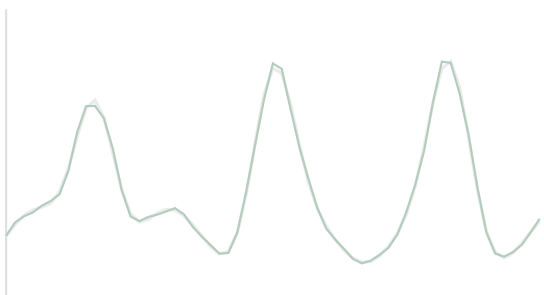


Illustrative prices for one site, [year].

[Logo] logo.svg







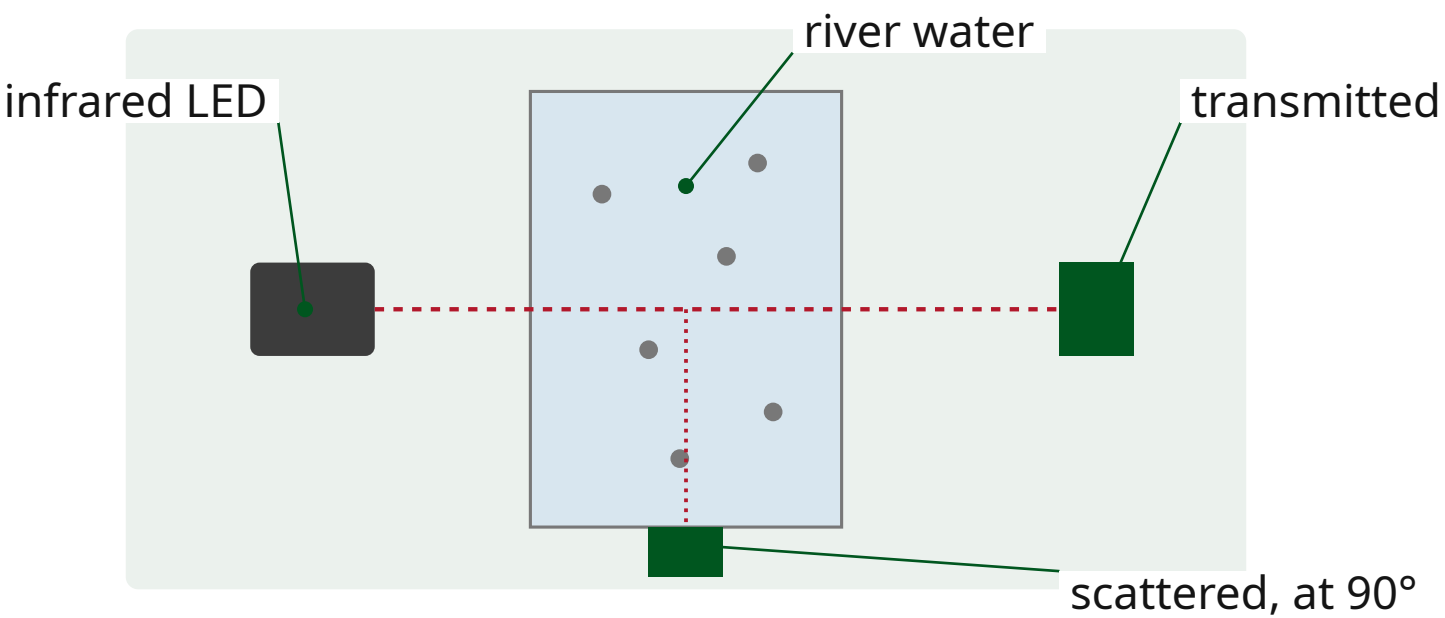
1. Need

2. Sensor

3. Field test

Optics

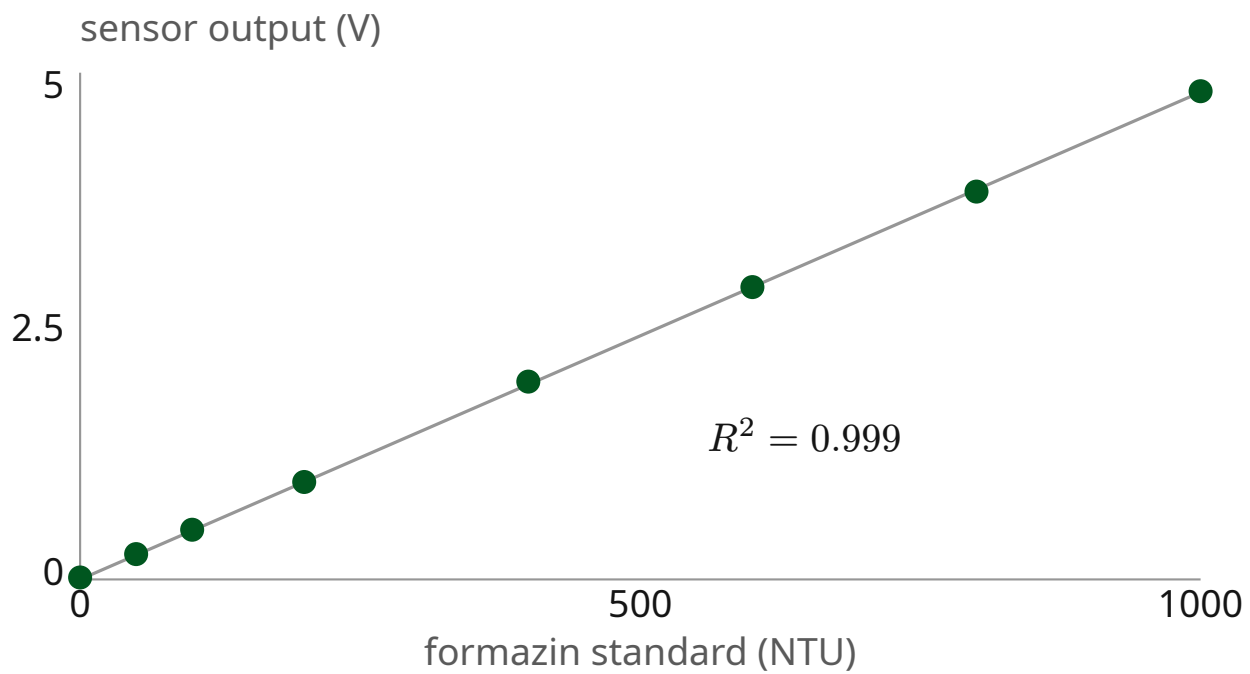
An infrared LED and two photodiodes measure scattered and transmitted light



Calibration

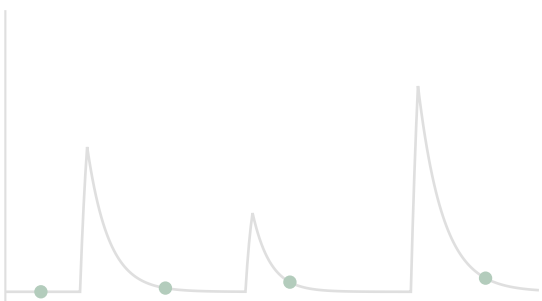
[Logo] `logo.svg`

The output is linear against formazin standards up to 1000 NTU

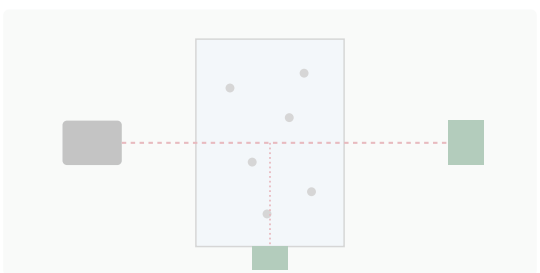


Illustrative calibration, eight standards.

[Logo] logo.svg



1. Need



2. Sensor

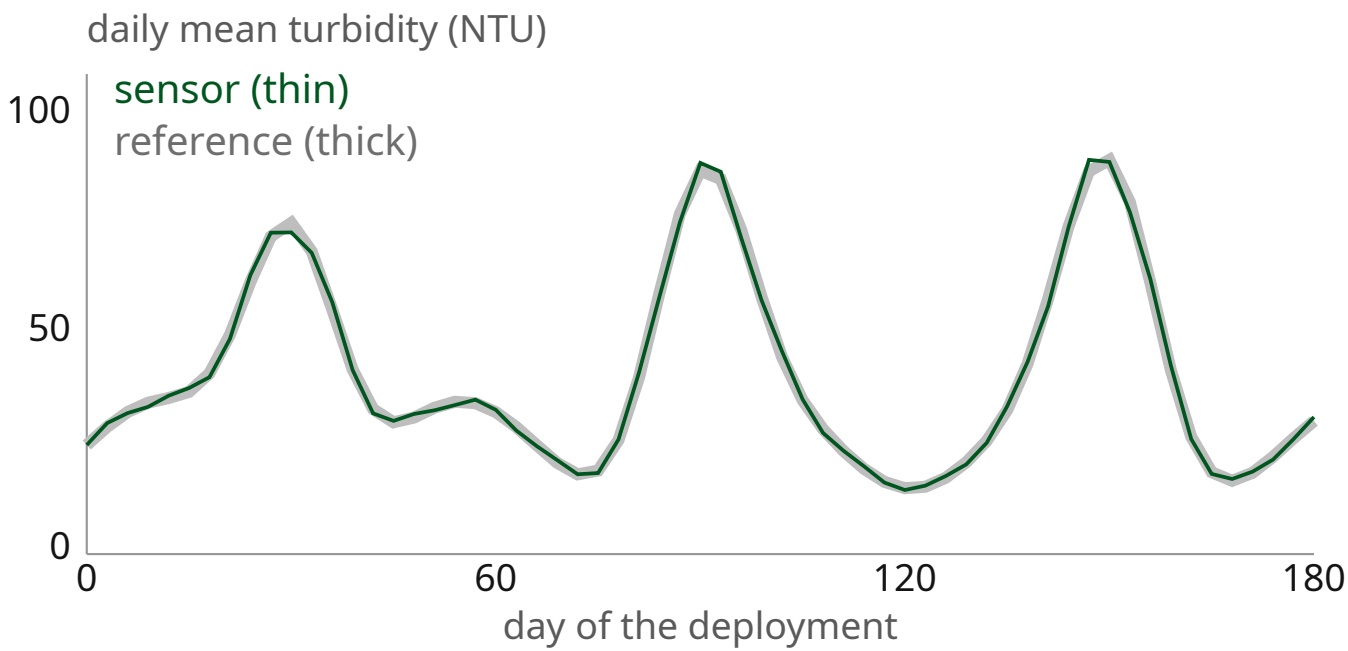


3. Field test

Agreement

[Logo] `logo.svg`

Over six months, daily means stayed within 5 % of the reference

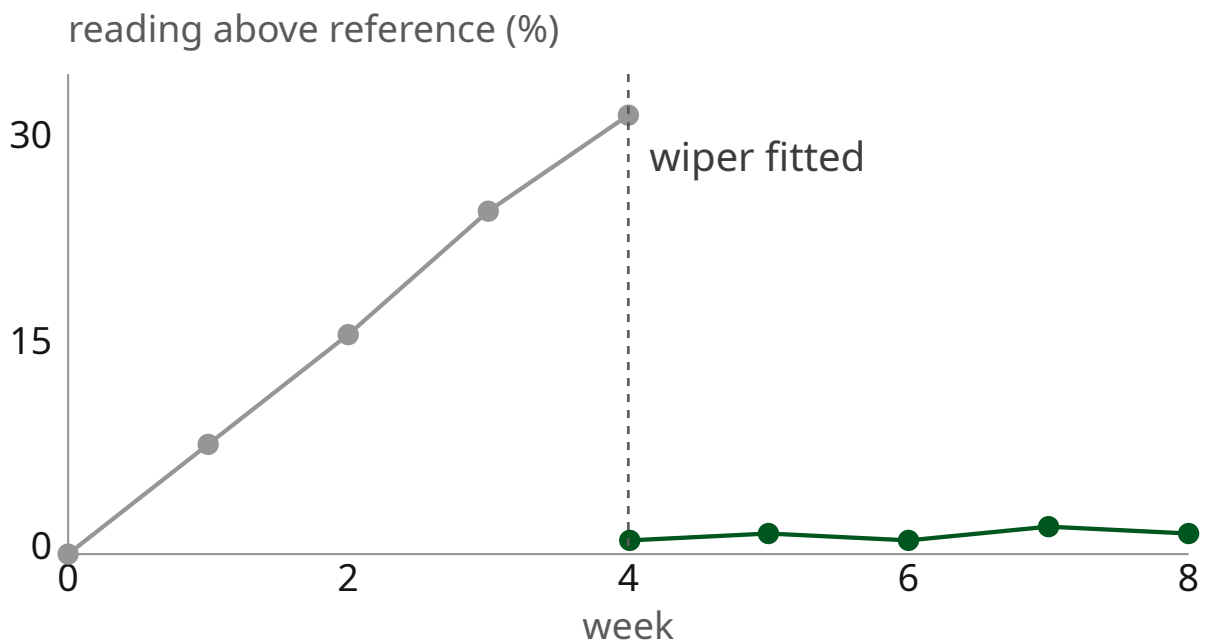


Illustrative record, [River name], [dates].

Fouling

[Logo] logo.svg

Algae raised readings by 8 % a week until a wiper cleared the window

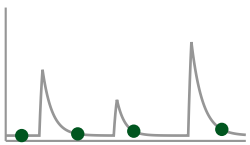


Illustrative data, first deployment.

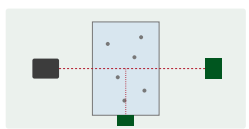
[Logo] logo.svg

Conclusion

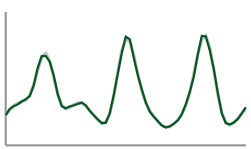
In summary, a $120 sensor records storm-driven turbidity within 5 % of a laboratory instrument



1. Weekly samples miss the storms that carry most sediment



2. Scattered and transmitted light give a linear, drift-free reading



3. With a wiper, six months stayed within 5 % of the reference

[Photo]

[Group name] · [email@example.org]

Questions?

Power budget

[Logo] logo.svg

The wiper uses 2 % of the battery budget

Load	Energy per day (Wh)	Share
Optics and logger	1.90	93 %
Radio	0.10	5 %
Wiper	0.04	2 %

Illustrative values.